

MOHAN M

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PROFESSIONAL SUMMARY

Cyber Security graduate (B.E., May 2026) with hands-on experience across Machine Learning, data analytics, and network security. Built two end-to-end ML projects — a real-time explainable network anomaly detection system and a sales forecasting platform — and completed a self-directed data analytics internship where cleaning and automation work reduced data quality issues by 40% and supported an 84%-accuracy churn model. IBM- and Infosys Springboard-certified. Seeking entry-level roles in Machine Learning Engineering, Data Science, Data Analytics, or Cybersecurity.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: Feature Engineering, Data Preprocessing, Model Training & Evaluation, Anomaly Detection, Time-Series Forecasting, Explainable AI (SHAP), Regression & Classification

Libraries & Tools: Pytorch, Scikit-learn, Pandas, NumPy, Matplotlib, Scapy, Streamlit, SHAP, Power BI

Cyber Security: Network Security, Vulnerability Assessment, Packet Analysis, Intrusion Detection, SOC Fundamentals

Dev Tools: Git/GitHub, VS Code, Excel, Jupyter Notebook

EXPERIENCE

Data Analytics Intern — NoviTech (The Innovation Partner), Tamil Nadu, India *May2025 – July2025*

- Reduced data quality issues by approximately 40% across 3 real-world business datasets (10K–50K rows) by designing standardized cleaning, preprocessing, and EDA workflows in Python, Pandas, and SQL
- Built 5+ interactive dashboards and automated reporting pipelines in Power BI and Excel, converting raw data into weekly, decision-ready insights for stakeholders
- Applied regression and classification models to a real customer dataset, achieving 84% model accuracy and enabling the team to prioritize high-risk accounts for retention outreach
- Completed a 30-day Data Analytics training program covering end-to-end analytics workflows (Certificate ID: NT_B20DA161)

PROJECTS

Real-Time Explainable Network Traffic Anomaly Detection — Academic Project *2025 – 2026*

Stack: Python, Scapy, Scikit-learn, TensorFlow, SHAP, Streamlit

- Engineered a hybrid ML framework combining an unsupervised Autoencoder with a supervised XGBoost classifier for real-time network intrusion detection
- Processed and feature-engineered the UNSW-NB15 dataset (2.5M records) with live packet capture via Scapy and Npcap
- Reduced false positives versus a single-model baseline by switching from CICIDS2017 to UNSW-NB15 and retuning the preprocessing pipeline
- Integrated SHAP explainability to surface top contributing features per prediction; deployed a live interactive Streamlit dashboard
- Repository: github.com/mohanm440

AI-Powered Sales Analytics & Demand Forecasting Platform — Personal Project *2026*

Stack: Python, Scikit-learn, Pandas, Matplotlib

- Built an end-to-end ML pipeline on real-world e-commerce data — from raw ingestion through feature engineering to supervised forecasting models
- Delivered a sales forecasting model using regression techniques, evaluated with MAE and RMSE
- Built interactive dashboards surfacing demand-planning and inventory insights for business decision-making

EDUCATION

Bachelor of Engineering, Cyber Security — Paavai Engineering College, Namakkal, Tamil Nadu *Nov 2022 – May 2026*

Relevant Coursework: Network Security, Machine Learning, Data Structures & Algorithms

Higher Secondary (Class XII) — The Gugai Higher Secondary School, Salem, Tamil Nadu *April 2022*

Class XII: 80% | SSLC (Class X): 81%

CERTIFICATIONS

- Python — IBM
- Machine Learning and Deep Learning — Infosys Springboard
- Data Analytics MasterClass — NoviTech R&D (Issued June 2025)
- Developing AI Applications with Python and Flask — Coursera (Issued July 2025)